

Industrial Internship Report on
ServiceHub – A Smart Service Marketplace Platform

Prepared by -

Shaleen Jaiswal

BCA

Aryan College, Ajmer

Industrial Internship

Upskill Campus × UniConverge Technologies Pvt. Ltd.

Executive Summary

This report presents the work completed during the Industrial Internship organized by **Upskill Campus** and **The IoT Academy** in collaboration with **UniConverge Technologies Pvt. Ltd. (UCT)**. The internship was based on an industry-oriented project that was to be completed, along with its documentation, within **6 weeks**.

My project, **ServiceHub – A Smart Service Marketplace Platform**, is a full-stack web application that enables customers to find and book services while allowing service providers to manage their offerings through dedicated dashboards. The project was developed using **Java Spring Boot, MySQL, HTML, CSS, Bootstrap, JavaScript, and JWT Authentication**.

This internship gave me valuable exposure to real-world software development and helped me improve my technical, problem-solving, and project development skills. Overall, it was a great learning experience that enhanced my understanding of full-stack development.

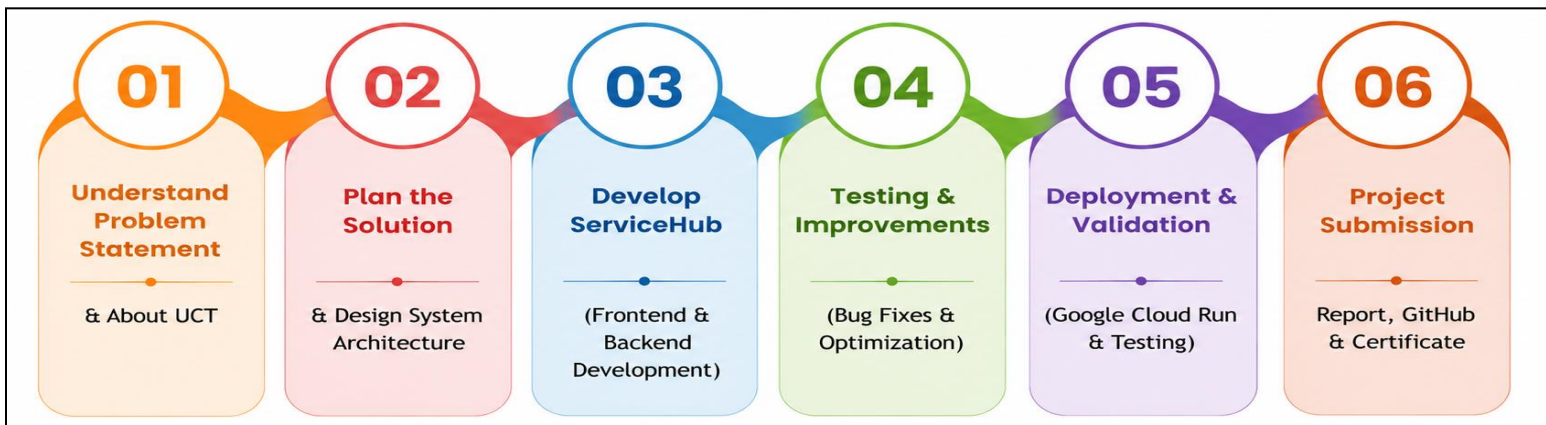
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1 Preface

Industrial internships provide students with an excellent opportunity to bridge the gap between academic knowledge and real-world industry practices. They help in developing technical expertise, problem-solving abilities, and a professional approach to software development by working on practical projects.

I successfully completed a 6-week Industrial Internship organized by Upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. (UCT). During this internship, I developed "ServiceHub – A Smart Service Marketplace Platform," a full-stack web application that enables customers to discover and book services while allowing service providers and administrators to efficiently manage the platform.



The internship followed a well-structured learning approach, including technical sessions, assignments, quizzes, and project milestones. Throughout the program, I gained practical experience in **Java Spring Boot, MySQL, HTML, CSS, Bootstrap, JavaScript, JWT Authentication, Git & GitHub**, and **Google Cloud Run** deployment.

This experience strengthened my understanding of the complete software development lifecycle—from planning and development to testing, deployment, and documentation.

I sincerely thank **Upskill Campus, The IoT Academy, UniConverge Technologies Pvt. Ltd. (UCT)**, my mentors, and my college for their continuous guidance and support throughout this internship. I also express my gratitude to my family and friends for their encouragement.

This internship has been a valuable learning experience that enhanced my technical skills, confidence, and readiness for future opportunities in the software industry.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



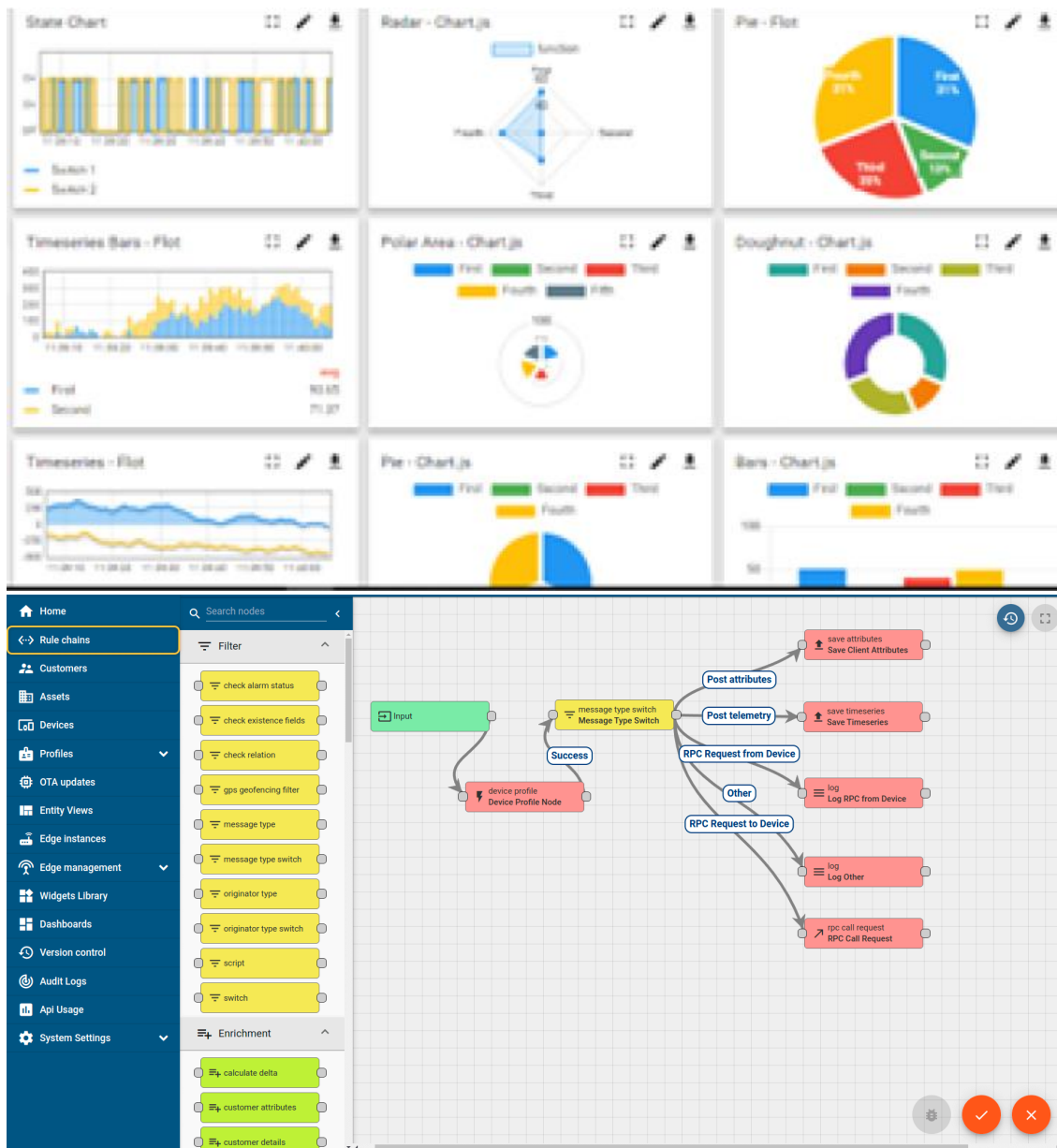
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (**FACTORY WATCH**)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

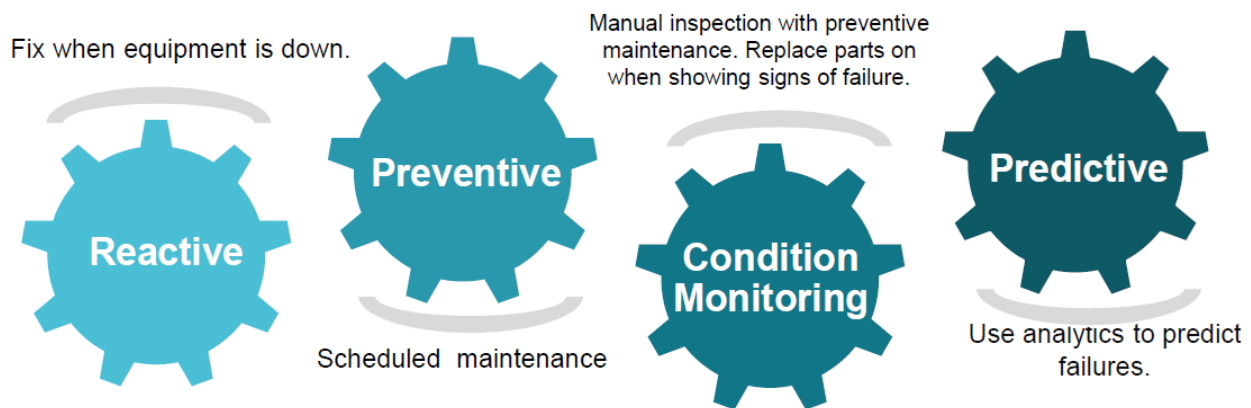


iii. **LoRaWAN™** based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

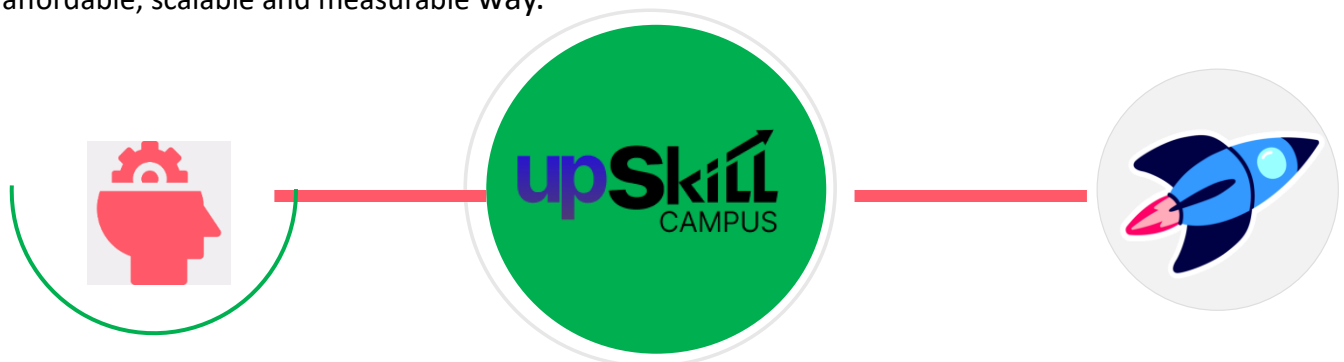
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

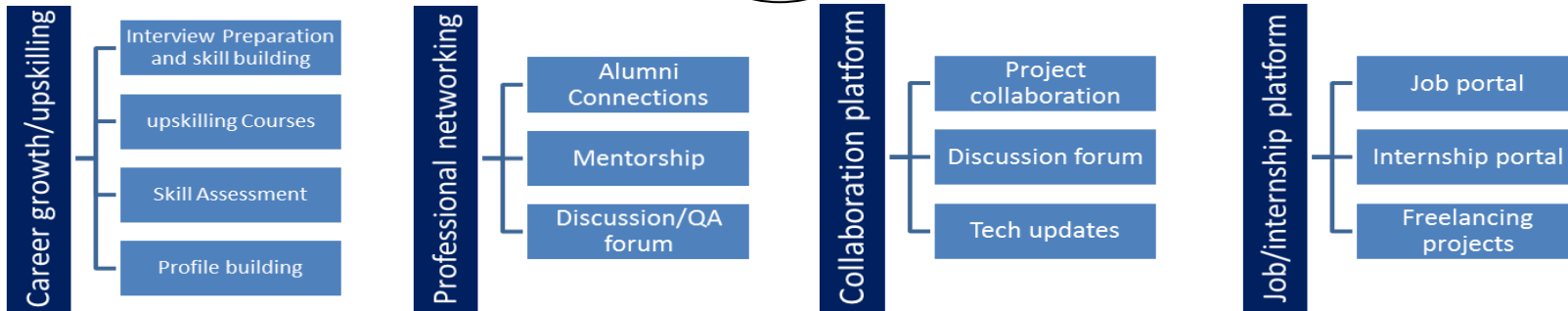
upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ▣ get practical experience of working in the industry.
- ▣ to solve real world problems.
- ▣ to have improved job prospects.
- ▣ to have Improved understanding of our field and its applications.
- ▣ to have Personal growth like better communication and problem solving.

2.5 Reference

Spring Boot Documentation

<https://spring.io/projects/spring-boot>

MySQL Documentation

<https://dev.mysql.com/doc/>

Bootstrap Documentation

<https://getbootstrap.com/docs/>

2.6 Glossary

Acronym	Full Form	Description
API	Application Programming Interface	Enables communication between frontend and backend.
JWT	JSON Web Token	Secure token used for authentication and authorization.
CRUD	Create, Read, Update, Delete	Basic operations performed on data in a database.
MVC	Model-View-Controller	Architecture pattern separating Model, View, and Controller.

3 Problem Statement

In today's digital world, customers often face difficulties in finding reliable service providers for their day-to-day needs, such as plumbing, electrical work, home cleaning, beauty services, and repairs. Most local service providers rely on phone calls, social media, or manual booking methods, making the process time-consuming, unorganized, and less transparent. Customers also lack a centralized platform to compare services, prices, ratings, and availability before making a booking.

Similarly, service providers face challenges in managing customer requests, maintaining booking records, and promoting their services to a wider audience. Manual management of appointments often leads to scheduling conflicts, missed bookings, and poor customer experience. Additionally, administrators do not have an efficient system to monitor users, services, and platform activities.

To address these challenges, **ServiceHub – A Smart Service Marketplace Platform** has been developed as a centralized web-based solution. The platform enables customers to discover and book services easily, allows merchants to manage their service listings and bookings through dedicated dashboards, and provides administrators with complete control over users, categories, services, and reviews. The application ensures secure authentication, efficient booking management, and a responsive user experience using modern full-stack web technologies.

The primary goal of this project is to simplify service booking, improve communication between customers and service providers, and provide a secure, scalable, and user-friendly digital marketplace.

4 Existing and Proposed solution

➤ Existing Solution

Currently, many local service providers rely on manual methods such as phone calls, social media platforms, or personal contacts to receive customer requests. Customers often spend considerable time searching for reliable providers, comparing services, and scheduling appointments. These traditional methods lack transparency, centralized management, and efficient booking mechanisms.

- **Limitations of the Existing System**
- Manual booking and scheduling process.
- No centralized platform for multiple service providers.
- Difficulty in comparing services and prices.
- Limited tracking of bookings and customer history.
- Lack of secure authentication and user management.
- Inefficient communication between customers and providers.

➤ Proposed Solution

To overcome these limitations, **ServiceHub – A Smart Service Marketplace Platform** has been developed as a centralized web application that connects customers, service providers, and administrators on a single platform.

The system enables customers to search and book services, merchants to manage their service listings and bookings, and administrators to monitor platform activities through dedicated dashboards. Secure authentication, role-based access control, responsive design, and efficient database management ensure a reliable and user-friendly experience.

- **Key Features**

- Secure User Authentication (JWT)
- Customer Dashboard
- Merchant Dashboard
- Admin Dashboard
- Service Management
- Booking Management
- Categories & Reviews

- Responsive User Interface
- Cloud Deployment

4.1 Code submission (Github link)

GitHub Repository:

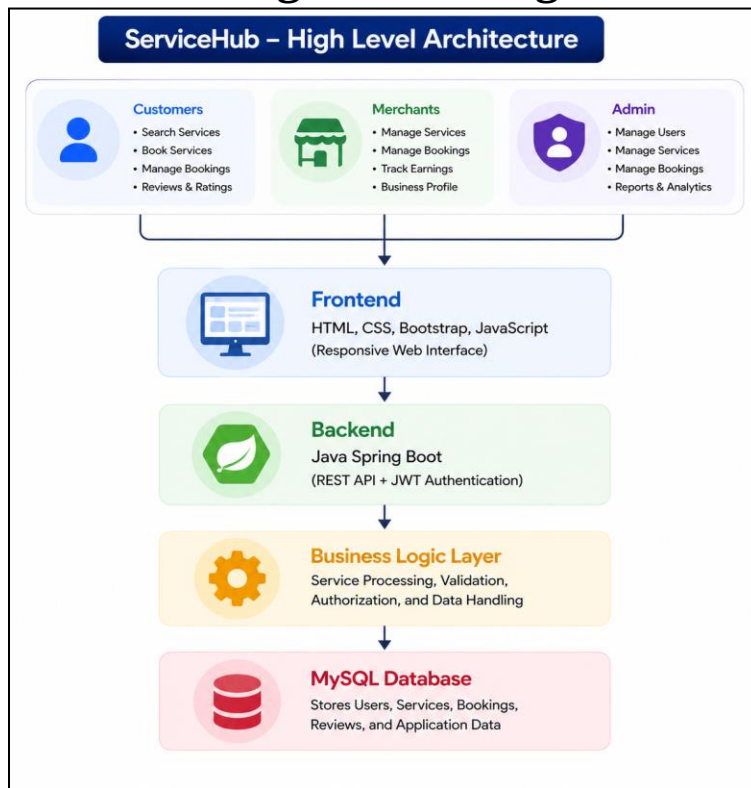
<https://github.com/Shaleen57551/upskillcampus>

4.2 Report submission (Github link) :

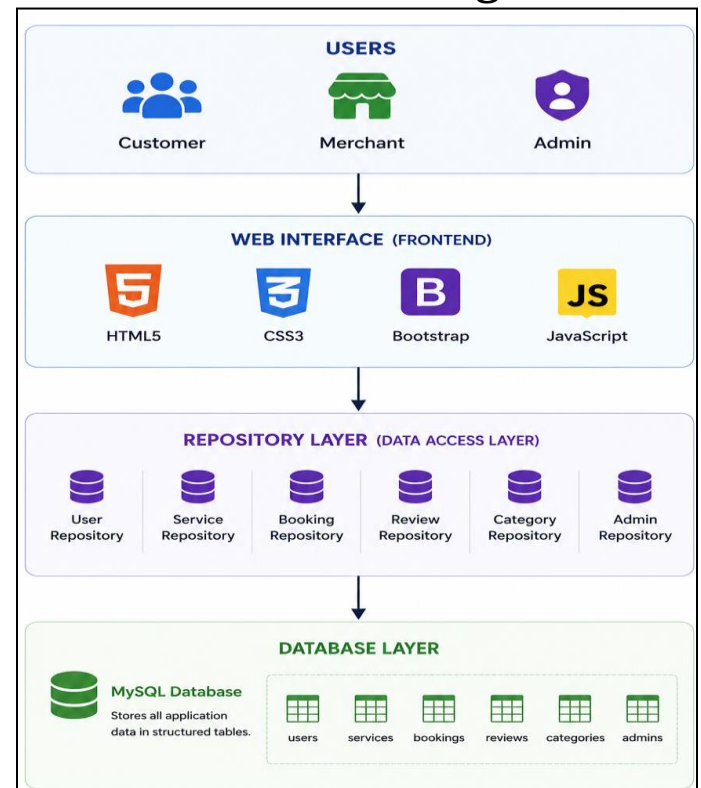
5 Proposed Design/ Model

The proposed solution, **ServiceHub – A Smart Service Marketplace Platform**, is designed as a secure, scalable, and user-friendly web application that connects customers with service providers through a centralized online platform. The application follows a **three-tier architecture**, consisting of the presentation layer, business logic layer, and database layer. It enables customers to search and book services, merchants to manage their offerings, and administrators to monitor and manage the platform efficiently. The system is developed using **Java Spring Boot**, **MySQL**, **HTML**, **CSS**, **Bootstrap**, and **JavaScript**, ensuring a responsive interface and secure backend services.

5.1 High Level Diagram



5.2 Low Level Diagram



5.2 Interfaces

The ServiceHub application provides a responsive and user-friendly interface for customers, merchants, and administrators. The system is designed using **HTML5, CSS3, Bootstrap 5, and JavaScript** to ensure smooth navigation and ease of use.

The major interfaces developed are:

- **Home Page**
- **Login Page**
- **Registration Page**
- **Customer Dashboard**
- **Merchant Dashboard**
- **Admin Dashboard**
- **Service Listing Page**
- **Booking Management Page**

6 Performance Test

Performance testing was conducted to evaluate the responsiveness, reliability, and stability of the ServiceHub application under normal operating conditions. The application was tested on different modules, including user authentication, service management, booking operations, and dashboard functionalities.

The results indicate that the application performs efficiently, providing quick response times and smooth navigation. The Spring Boot backend processes API requests effectively, while the MySQL database ensures reliable data storage and retrieval. The responsive frontend built with Bootstrap 5 delivers a consistent user experience across different devices.

The deployment on Google Cloud Run further improves application availability and accessibility through cloud hosting.

6.1 Test Cases

Test Case ID	Test Case	Expected Result	Status
TC001	User Registration	New user registered successfully	✓ Pass
TC002	User Login	User authenticated successfully	✓ Pass
TC003	Add Service	Service added successfully	✓ Pass
TC004	Book Service	Booking created successfully	✓ Pass
TC005	Dashboard Access	Correct dashboard displayed	✓ Pass
TC006	Review Submission	Review saved successfully	✓ Pass
TC007	Database Operations	Data stored and retrieved correctly	✓ Pass

6.1 Test Procedure

The application was tested by executing each module individually and verifying its functionality against the expected outcomes. User registration, login, service management, booking, review submission, and dashboard operations were performed using different user roles. Database consistency, API responses, and user interface behavior were also validated to ensure the system functioned correctly under normal operating conditions.

6.2 Performance Outcome

The testing confirmed that the **ServiceHub** application performs efficiently and reliably. All major functionalities operated successfully with accurate database

transactions and secure JWT-based authentication. The responsive user interface provided a smooth experience across different devices, while the deployment on **Google Cloud Run** ensured stable and accessible application performance.

7 My learnings

During the development of the **ServiceHub** project, I gained practical experience in full-stack web development and software engineering. I learned how to build a secure web application using **Java Spring Boot, MySQL, HTML, CSS, Bootstrap, and JavaScript**.

This project enhanced my understanding of **REST APIs, JWT-based authentication, database design, CRUD operations, and role-based access control**. I also learned how to integrate frontend and backend components, manage source code using **Git and GitHub**, and deploy a web application on **Google Cloud Run**.

Working on this project improved my problem-solving, debugging, and project management skills while giving me confidence in developing real-world web applications. The internship provided valuable hands-on experience and strengthened my foundation in modern full-stack development.

8 Future work scope

The **ServiceHub** platform can be further enhanced by introducing advanced features to improve functionality, scalability, and user experience. Future enhancements may include:

- Integration of secure online payment gateways.
- Real-time chat and notification system.
- AI-based service recommendations.
- GPS-based location tracking for nearby service providers.
- Mobile application for Android and iOS.
- Advanced analytics and reporting dashboards.
- Multi-language and multi-currency support.
- Integration with third-party APIs for maps and communication.

These enhancements will make the platform more efficient, scalable, and suitable for large-scale real-world deployment.